

DATA

SCIENCE

COURSE

FROM COLLEGE TO IT INDUSTRIES
CONCEPTS



Published By

TSAR IT PRIVATE LIMITED

CONCEPTS

1. INTRODUCTION TO DATA SCIENCE
2. PYTHON FOR DATA SCIENCE
3. STATISTICS & MATHEMATICS
4. DATA WRANGLING & EXPLORATORY DATA ANALYSIS (EDA)
5. MACHINE LEARNING (ML)
6. ADVANCED ML & MODEL DEPLOYMENT
7. BIG DATA & SQL
8. DATA VISUALIZATION TOOLS
9. INDUSTRIAL TOOLS & PRACTICES
10. REAL-WORLD PROJECTS
11. INTERVIEW PREPARATION



ABOUT OUR COMPANY

At **TSAR IT Private Limited**, we are a cutting-edge software company dedicated to providing innovative digital solutions to businesses and organizations. Located in Andhra Pradesh, we specialize in a range of services and software products tailored to meet the unique demands of our clients.

we like challenges to solve and give solutions to your business growth and you can monitor your business and employees work easily,

We offer services for a wide range of fields such as software development, web development, graphic development, cross platform development etc. For more information navigate to our services section.

“Explore The Services We Offer For You” we are experts in these services, we have strong experience and technical expert team, we have our own German Technology server, our team will complete and give you project in time.

we will provide our services like how clients feel comfortable, we can handle any type of project with low cost and we are expert in install and maintaining projects.

N RAKESH - CEO

Internship Courses:

In addition to our product offerings, **TSAR IT Private Limited** also provides Internship Courses for aspiring software developers, engineers, and tech enthusiasts. Our internship programs offer hands-on experience and training in cutting-edge technologies, including:

- **Java Development**
- **Spring Boot & React Full-Stack Development**
- **AI & Machine Learning**
- **Cloud Computing**
- **Software Testing & QA**

These internships are designed to bridge the gap between academic learning and real-world software development, preparing the next generation of IT professionals.

At **TSAR IT Private Limited**, we believe in the power of technology to drive success, innovation, and transformation. With a dedicated team of experts, we are committed to delivering high-quality, customized software solutions while fostering new talent through our internship programs.

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DATA SCIENCE COURSE SYLLABUS

1. INTRODUCTION TO DATA SCIENCE

- 1.1. What is Data Science?
- 1.2. Data Science Lifecycle
- 1.3. Applications in Industries (Healthcare, Finance, E-commerce, etc.)
- 1.4. Tools and Technologies Overview

2. PYTHON FOR DATA SCIENCE

- 2.1. Python Basics (Syntax, Data Structures)
- 2.2. NumPy for Numerical Computing
- 2.3. Pandas for Data Manipulation
- 2.4. Matplotlib & Seaborn for Data Visualization
- 2.5. OOPs Concepts in Python

3. STATISTICS & MATHEMATICS

- 3.1. Descriptive Statistics (Mean, Median, Variance)
- 3.2. Inferential Statistics (Hypothesis Testing, p-value)
- 3.3. Probability Distributions (Normal, Binomial, Poisson)
- 3.4. Linear Algebra (Vectors, Matrices)
- 3.5. Calculus Basics (Gradient, Derivatives)

4. DATA WRANGLING & EXPLORATORY DATA ANALYSIS (EDA)

- 4.1. Data Cleaning (Handling Missing Values, Outliers)
- 4.2. Feature Engineering
- 4.3. Data Transformation (Normalization, Standardization)
- 4.4. EDA Techniques (Univariate, Bivariate, Multivariate Analysis)
- 4.5. Correlation and Heatmaps

5. MACHINE LEARNING (ML)

5.1. Introduction to ML (Supervised vs. Unsupervised)

5.2. Supervised Learning:

- Linear & Logistic Regression
- Decision Trees, Random Forest
- SVM, k-Nearest Neighbors (k-NN)
- 5.3. Unsupervised Learning:
 - Clustering (k-Means, Hierarchical)
 - Dimensionality Reduction (PCA, t-SNE)
- 5.4. Model Evaluation (Accuracy, Precision, Recall, ROC-AUC)

6. ADVANCED ML & MODEL DEPLOYMENT

6.1. Ensemble Techniques (Bagging, Boosting – XGBoost, AdaBoost)

6.2. Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV)

6.3. ML Pipelines

6.4. Deploying Models with Flask/Django

6.5. Cloud Deployment (AWS SageMaker, GCP AI Platform)

7. BIG DATA & SQL

7.1. SQL Basics (Queries, Joins, Subqueries)

7.2. Advanced SQL (Window Functions, Indexing)

7.3. Big Data Concepts (Hadoop, Spark)

7.4. NoSQL Databases (MongoDB, Cassandra)

8. DATA VISUALIZATION TOOLS

8.1. Tableau (Dashboards, Stories)

8.2. Power BI

8.3. Advanced Python Visualization (Plotly, Bokeh)

9. INDUSTRIAL TOOLS & PRACTICES

- 99.1. Version Control (Git, GitHub)
- 9.2. CI/CD Pipelines (Jenkins)
- 9.3. Containerization (Docker, Kubernetes)
- 9.4. Agile & Scrum Methodologies

10. REAL-WORLD PROJECTS

- 10.1. Predictive Analytics (Sales Forecasting)
- 10.2. Customer Segmentation (Clustering)
- 10.3. Sentiment Analysis (NLP)
- 10.4. Recommendation Systems (Collaborative Filtering)
- 10.5. End-to-End Project Deployment

11. INTERVIEW PREPARATION

- 11.1. Data Science Case Studies
- 11.2. Mock Interviews
- 11.3. Resume Building

For Registration process:

Step 1:

Open your Browser & paste the link given below

Step 2:

Given Your Details

Step 3:

Click Register Button

REGISTRATION LINK :<https://internship.tsaritservices.com/registrationForm>

APPLY

For More Details:

TSAR IT PRIVATE LIMITED

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